

From one to zero: Comparator-Heated In-memory Lane Learning with Zero-Ended Noise

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We present CHILL ZEN—Comparator-Heated In-memory Lane Learning with Zero-Ended Noise—a generative architecture for probabilistic in-memory hardware. Images are represented by additive discrete codes. Locally taught lane banks generate a backbone code autoregressively, repair it, draw residual patch codes in parallel, and apply one joint repair sweep. Generation uses about 3600 noisy lane reads and 131 sequential read steps, without an equilibrium sampling phase. In the emulator, pooling suppresses device noise and an adjustable comparator-noise term supplies most of the randomness needed for generation. On binarized Fashion-MNIST, the two-level system scores FID 18.1 when trained on grayscale images and 9.7 when trained on binary images. Each score uses the best setting on its tested noise grid. Under the stated energy models, the two-level system uses about one eighth as much energy per image as the published eight-step DTM (2×10^{-9} versus 1.57×10^{-8} J). Its FID is also lower than the DTM’s reported 24.9. We use the released DTM replication pipeline for scoring. We have not verified the original paper’s exact scoring procedure, and the training splits differ. The energy comparison includes modeled readout and lane-decoder energy. We also estimate the energy of a selector-column design that eliminates the nominal crossbar’s sneak paths. The emulator does not model these paths. The energy advantage comes from fewer sampling operations per image: about 3600 noisy lane reads versus 10^7 conditional updates in the DTM chain.

I. INTRODUCTION

The energy consumption of generative AI has become a first-order constraint on its deployment, and it has prompted a search for computers in which physics performs the sampling [1]. Probabilistic bits built from stochastic nanomagnets and related devices have performed optimization and learning tasks at very low energy [2–4]. Jelinčič *et al.* [5] recently proposed a generative model and architecture using all-transistor probabilistic circuits, with an energy analysis.

That work states the central obstacle cleanly: the mixing–expressivity tradeoff. As a monolithic energy-based model (EBM) fits multimodal data better, energy barriers grow between the modes, and the mixing time of any locally informed sampler grows with them. The Denoising Thermodynamic Model (DTM) chains $T = 2$ –8 EBMs to reverse a noising process, as diffusion models do [6, 7], keeping each step easy to mix. The Denoising Thermodynamic Computer Architecture (DTCA) implements these steps with sparse bipartite Boltzmann machines. Each node is a subthreshold-transistor random-number generator (RNG) with a measured decorrelation time of $\tau_0 \approx 100$ ns [5, 8].

Each read instruction specifies its sampling temperature. The emulator also allows noiseless reads. Generation uses one hot pass over the code followed by one cold sweep. We call the system CHILL ZEN, for Comparator-Heated In-memory Lane Learning with Zero-Ended Noise.

Pooling weight pairs on a lane reduces read noise to the output node’s kT/C floor. At 10 mV, the deployed lanes supply less than 40% of the required draw noise on the first read and less than 10% thereafter, given the stated

read-voltage and offset assumptions. The periphery must therefore supply the sampling temperature. Each lane has a comparator with adjustable input-referred noise. At the operating point, it supplies most of the draw’s variance, exceeding 90% after the first read. The pooled devices supply the rest. In this respect CHILL ZEN and the DTCA are alike: each obtains its randomness from a noise circuit at every node. One CHILL ZEN image needs about 3600 noisy reads; an eight-step DTM chain needs of order 10^7 random conditional updates. These operation counts determine the energy accounting of Sec. VB. We state the comparator’s design requirement and measure the operating point.

Our contributions are:

1. CHILL ZEN, a generative model class for lane hardware, defined by its code, its paired generator and repair banks, and its one-pass, one-sweep procedure (Sec. III);
2. the two-level result on grayscale Fashion-MNIST [9], with reconstruction controls (Secs. IV B and IV D);
3. the comparator-noise result: pooled reads quench their own noise, the comparator supplies the draw temperature, and a sweep of its noise setting fixes the operating point and the design requirement (Sec. IV C);
4. a comparison with the DTM/DTCA as design points, a modeled energy accounting, and a head-to-head on binarized Fashion-MNIST scored with the released evaluation code of Ref. [5] (Sec. V).

Every result is emulated lane computation: device models with the read-noise physics of Sec. II, every read and

adapt issued as a lane instruction, and decode, render, and judging in host-side floating point.

II. THE SUBSTRATE

A. The kT-bit and its read noise

The storage and sampling element is a differential pair of metastable memristors read through a voltage divider. With conductances G_a and G_b and read voltage V , the divider reports

$$V_y = V \frac{G_a - G_b}{G_a + G_b}, \quad y = \frac{V_y}{V} \in [-1, 1], \quad (1)$$

and the pair carries two state variables,

$$w = \frac{G_a - G_b}{G_a + G_b}, \quad m = G_a + G_b. \quad (2)$$

The weight w is the balance of the pair and the magnitude m is its total. The magnitude controls the pair's response to feedback: a fixed conductance increment produces a smaller change in w at larger m . For forward-only increments, conductances measured in units of the increment can be interpreted as Beta pseudo-counts a and b , with

$$\mu = \frac{a}{a+b}, \quad \kappa = a+b, \quad \sigma^2 = \frac{\mu(1-\mu)}{\kappa+1}. \quad (3)$$

The deployed rules also use reverse feedback, so magnitude need not grow monotonically and is not a cumulative evidence count. The Beta analogy describes forward accumulation; it does not make the trained pair a Bayesian posterior or establish a $1/n$ learning schedule for the full rule.

A read returns the stored weight plus noise. The total noise variance adds the thermal and flicker contributions from the devices and the contribution from the read comparator:

$$\begin{aligned} \sigma_{\text{thermal}} &\propto \frac{\sqrt{k_B T}}{V\sqrt{m}}, & \sigma_{\text{flicker}} &\propto \frac{1-w^2}{\sqrt{m}}, \\ \sigma_{\text{comparator}} &= \frac{v_n}{V}, \\ \sigma_y &= \sqrt{\sigma_{\text{thermal}}^2 + \sigma_{\text{flicker}}^2 + \sigma_{\text{comparator}}^2}. \end{aligned} \quad (4)$$

The thermal term scales as $1/V$, so each instruction can adjust temperature through the read voltage. At full read voltage, device noise approaches the flicker floor (Fig. 1). We call such a read cold and treat it as $T = 0$. Section IV C tests this simplification. At fixed pulse width, both device terms fall as $1/\sqrt{m}$. The flicker term also includes $(1-w^2)$, so it falls as w approaches ± 1 . These trends follow the posterior in Eq. (3): uncertainty falls with more evidence and near either endpoint. We use a settling-limited pulse, $t = 5C/G_\Sigma$. The pooled conductance then cancels from the thermal term, leaving the

output node's kT/C noise: $\sigma_{\text{thermal}}^2 \approx k_B T / (C_{\text{node}} V^2)$. Pooling still reduces thermal noise because the added wiring increases node capacitance in proportion to the number of pairs (Appendix B, $C_{\text{node}} \propto K$). The sign of one subthreshold read is a probabilistic bit with $P(+) = \Phi(w/\sigma_y)$ whose bias is the stored weight.

The comparator term is different in kind. This term is the comparator's input-referred noise v_n divided by the read voltage. It does not depend on w or m . That noise is kT/C noise on the regeneration nodes plus the preamplifier's own thermal noise [10]. Circuit settings control this noise. Reducing preamplifier bias or adding input noise increases it. In a clocked comparator, triggering the latch earlier also increases it. It falls only at a cost in area and power, down to a floor set by the comparator's design class. We model comparator noise as a trim-register setting specified by each read instruction, alongside voltage and pulse width. Each read thus sets its own sampling temperature. The comparator's input offset is a static bias caused by transistor mismatch [11]. A one-time trim corrects it for each lane. Section IV C gives the allowed offset.

Pooling reduces device noise, leaving the comparator to supply most of the draw noise (Sec. IV C). Each lane read pools the magnitudes of the selected pairs. In a well-trained pool, accumulated evidence reduces flicker noise, while the added node capacitance reduces kT/C thermal noise. What is left is a floor set by the read voltage and the node capacitance. At 10 mV, the model gives lane noise of about 0.02–0.04 without the comparator. The geometry in Appendix B gives up to 0.07 on the first read, when node capacitance is about 12 fF. The draw needs at least 0.1. Supplying that from the node alone would require about a 1 mV read, below the voltage assumed by the offset budget. The comparator supplies the remaining noise. CHILL ZEN sets the generation temperature by adjusting each lane's preamplifier bias: low bias for hot reads, full bias for cold reads.

Prior work has used device variability for sampling. Dalgaty *et al.* [12] run Markov chain Monte Carlo in a 16,384-device array by treating the cycle-to-cycle variability of a memristive write as the proposal distribution, and Cai *et al.* [13] use intrinsic device noise as the annealing source in a memristor Hopfield network. Both draw on a single device's variability. In Eq. (4) the device is one of three noise sources. The read voltage is another: the same weight reads hot or cold. The comparator is the third, and once hundreds of pairs are pooled on a lane it is the largest term.

B. The neural lane

A neural lane is an array of unit-crossbar pairs pooled on a shared 2-1 divider. By Kirchhoff's laws the selected

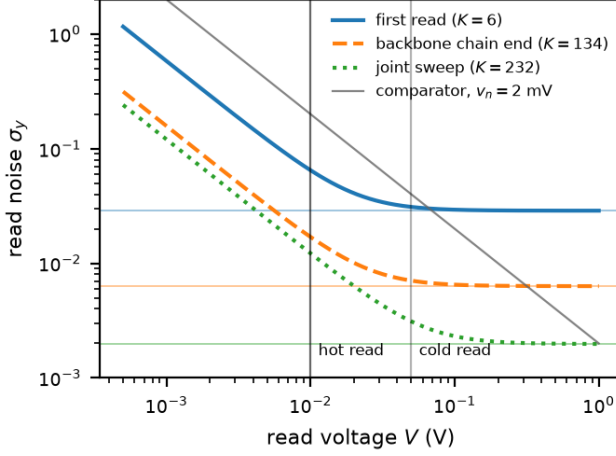


FIG. 1. **The devices alone set only the floor of a read’s noise, and pooling lowers that floor.** Read noise against read voltage at the settling-limited pulse of Sec. IV C. The thermal term is the kT/C noise of the lane’s output node, from the geometry of Appendix B plus 5 fF of comparator input, for the first read of the chain ($K = 6$), the end of the backbone chain ($K = 134$), and the joint sweep ($K = 232$); the thin horizontal lines are the flicker floors at each pool’s measured magnitude. Node capacitance grows with the number of pooled pairs, reducing noise by up to a factor of five relative to the first read. The comparator’s operating level, $v_n = 2$ mV, is drawn for scale: at the 10 mV hot read it is the larger term on every read.

pairs report

$$y = \frac{\sum_{i \in A} (G_{a,i} - G_{b,i})}{\sum_{i \in A} (G_{a,i} + G_{b,i})} = \frac{\sum_{i \in A} m_i w_i}{\sum_{i \in A} m_i}, \quad (5)$$

the magnitude-weighted average of the selected weights. Selection is an Activation Address Tuple (AAT): an ordered tuple of integers, one per address space. Each space influences a read in proportion to its share of the pooled magnitude, roughly $1/K$ for K active spaces. Addresses without measured values therefore dilute those with measured values. Conditioning addresses must be repeated across several spaces to have enough influence. These constraints determine the model class.

Instruction codes in the kT-RAM instruction set (Table I) are two letters: the first is the drive direction (F forward, R reverse), the second the operation. Forward feedback accumulates magnitude and reverse feedback depletes it; every working routine pairs a forward operation with a reverse one.

C. WTA Groups and the lane routines

Lanes form winner-take-all (WTA) Groups, with one lane per address-space symbol. This paper uses three routines, described in detail in an online chapter series [14]. In the *supervised three-case rule*, all lanes of a group

TABLE I. The kT-RAM instruction set.

Instruction	Kind	Effect
FF, RF	read, full voltage	return y ; the drive adapts the pair (repeated FF anti-Hebbian, RF Hebbian)
FFLV, RFLV	read, subthreshold	return y with zero disturbance
FH/RH, FL/RL	supervised feedback	drive w up / down
FU/RU, FA/RA	unsupervised feedback	reinforce / oppose the current sign
FZ, RZ	zero feedback	no push on w ; m grows / shrinks

are read with FF; the labeled lane receives RH; a wrong lane that fired receives RL (hard) or nothing beyond the paired reverse read (soft); a correctly silent lane receives RF. Cold argmax readout of a group taught this way is a classifier at parity with logistic regression on the identical encoding (Iris: 0.962 against 0.961 over twenty splits). Hard feedback creates a margin between the winner and the other lanes. Soft feedback retains all outcomes supported by the data, so a hot read samples from a distribution defined by the learned lane scores. Generation runs as sequential lane reads: clamp what is known, read one space, commit, continue. *RankCut* ranks a group’s above-threshold lanes using a shared falling reference and one comparator per lane. It requires no per-lane analog-to-digital converter.

D. The emulator

All results are emulated lane computation on *ktram-neural-core* [15]: device models with the read-noise physics above, a torch runtime bit-exact against the reference core, and every read and adapt issued as a lane instruction. The emulator’s crossbar fidelity is ideal: a read returns the selected pair of each space alone, with no contribution from the unselected devices of the unit crossbar (Limitation 2). The physical memristors and crossbars exist as fabricated and commercially sold hardware [16]; the lane-scale system does not yet. The emulator’s device noise gain stays fixed at its reference value of 0.02, including during hot reads. A hot read is driven by the read voltage and by the comparator term of Eq. (4), both physical operating points.

III. THE MODEL CLASS

A. The code

We encode images with per-position dropout additive quantization (dropout-AQ): k books of S atoms, with keep probability p . This approach builds on vector and additive quantization [17–19] and on generation from dis-

crete codes [20, 21]. A code is a point in $\mathbb{A}_S^k$ (k address spaces of S symbols each), mirroring the codeword space $\mathbb{F}_q^n$ of coding theory. The training objective marginalizes Bernoulli address deletion,

$$\mathbb{E}_z \left\| x - b - \sum_j z_j a_j \right\|^2 = \left\| x - b - p \sum_j a_j \right\|^2 + p(1-p) \sum_j \|a_j\|^2, \quad (6)$$

with $z_j \sim \text{Bernoulli}(p)$. Both partial and complete codes use the same fixed linear decoder:

$$\hat{x} = b + p \sum_{j \in \text{present}} a_j. \quad (7)$$

The codebooks are fitted on the host by alternating discrete encoding and a joint double-precision ridge solve. At $p = 1$, the objective is standard additive quantization [19]. At $p < 1$, training also includes partial codes, though some may still decode to incoherent images.

Stripped of the hardware, the model class is generation over a linear additive-quantization codebook with masked-context training and a parallel reconciliation sweep, which places it beside MaskGIT [21] and the VQ-VAE lineage [20]. The fixed linear decoder handles partial codes and can run as lane reads, whose energy is counted in Sec. VB. Each level pairs a soft bank for sampling with a hard bank for repair, both trained through local lane instructions. A masked transformer uses one network and a confidence schedule.

B. Paired banks and the generation procedure

Each CHILL ZEN level pairs a soft generator bank G with a hard repair bank R. G uses hot reads. R trains on corrupted pools and uses one cold parallel sweep to commit all updates. The two-level system evaluated here has a *backbone* (level 1: a global code over the whole canvas, with banks G1 and R1) and a *patch level* (level 2: a residual code on a grid of patch slots, with bank G2 and a cross-level joint repair bank). The *render* is the backbone decode plus the patch decode. For context $c = (z_{<m}, \ell)$, the draw is

$$z_m = \arg \max_s \{y_{m,s}(c) + \epsilon_{m,s}\}, \quad \epsilon_{m,s} \sim \mathcal{N}(0, \sigma_{m,s}^2), \quad (8)$$

with independent noise in the emulator. This defines a conditional choice probability for each symbol; the teaching rule does not calibrate it to the data conditional. Generation runs as follows (Fig. 2):

1. Clamp the label, replicated across six address spaces.
2. Backbone draw: read the 128 global books autoregressively, choosing the lane with the highest noise-perturbed score at each step, then apply one cold

R1 sweep. Hot reads use $(V, v_n) = (10 \text{ mV}, 2 \text{ mV})$, the best critic setting on the tested grid (Sec. IV C). Section VC reports FID at its own best setting on that grid.

3. Patch level: one parallel hot G2 read, at the same operating point, over all 49 residual patch slots under the held global code.
4. One cold cross-level joint sweep reconciles both levels: all 226 addresses are targets, and the sweep may flip globals bottom-up and patches top-down in the same parallel read.
5. Decode and render: the fixed linear map of Eq. (7), run in host floating point in this paper’s emulation and counted in Sec. VB as the lane read it is designed to be.

Generation takes 131 sequential steps per image: 128 autoregressive reads and three parallel sweeps. Running the step 5 decoder on lanes adds one parallel read. The temperature schedule has two values. Every read uses the same read voltage, so the temperature is the comparator’s noise setting alone: $v_n = 2 \text{ mV}$ for draws and $v_n \rightarrow 0$ for decisions. Both settings occur within one image. The joint sweep runs once by design, because repeating it degrades the image. Each further cold dose lowers the critic by about 0.01, from 0.90 after one dose to 0.81 after eight, and pulls diversity and tone toward the class prototypes. Real encoded codes drift the same way under repeated doses. The sweep’s fixed points are the class prototypes, and the data are not among them. Warming the sweep does not help. With read noise at a tenth of the read voltage the critic stops decaying over doses and holds near 0.85, but diversity still falls, from 0.63 to 0.56 by the eighth dose. At the draw’s own operating point the sweep no longer functions: it flips about 200 of the 226 addresses on every dose and the critic drops below 0.5 after the first.

IV. TRAINING AND RESULTS

A. Setup and metrics

The benchmark is grayscale Fashion-MNIST [9], split 68 000 train and 2 000 evaluation. We report four metrics. *Critic* measures label agreement using a frozen pixel-space multilayer perceptron ($784 \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 10$); the target is its score on real images. *Div* measures within-class diversity as the fraction of pixel pairs differing by more than $1/16$. *Tone* measures the within-class spread of each image’s mean foreground level. *Std-ratio* measures foreground contrast relative to real images, with an alarm threshold of 1.10. Section VC reports FID on binarized renders using the DTM replication protocol.

The banks are trained using local lane instructions without gradients. The codec is fitted on the host, and

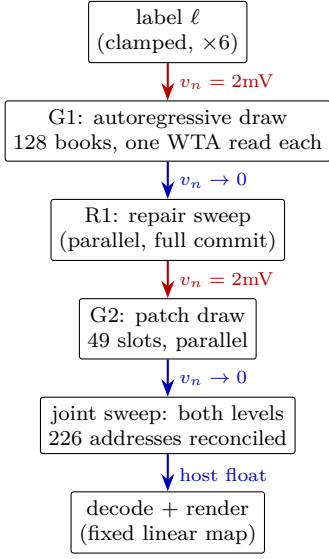


FIG. 2. **One CHILL ZEN generation uses both temperature regimes in turn.** Hot reads (red) draw the code and cold sweeps (blue) reconcile it; exactly two arrows are hot, both at the operating point $(V, v_n) = (10\text{ mV}, 2\text{ mV})$ of Sec. IV C.

the frozen critic is trained with Adam. Soft G banks train on Bernoulli-masked codes with labels retained. Hard R banks train on pools corrupted by marginal sampling and G fills. The joint bank trains on corrupted two-level states. Each bank is probed per epoch on the same 2000-image evaluation partition and the best-probe snapshot is kept. That partition also informed the fill-noise and generation-noise choices; it is a validation set, not an untouched test set. New generation seeds measure sampling variation for the selected banks, not uncertainty from training or model selection.

Repair-training fills use the 10 mV read with comparator register code 245 ($v_n = 5\text{ mV}$). This setting was chosen during development and carried over to the binary-trained stack. These runs cannot isolate the effect of fill noise: generation noise also varied, and the intermediate fill-sweep logs were not retained. The released scripts reproduce the final recipe. Two choices in that recipe, holding the label and committing a single cold dose, follow the pool-share argument of Sec. II B.

B. The two-level result

The deployed system is a 904-bit code plus the clamped label: a 128×16 global backbone at keep- p 0.5 (512 bits) and a 2×16 residual patch code on a 7×7 grid of 4×4 -pixel slots (392 bits). Table II reports means and seed ranges (max – min) over three generation seeds at $n = 1000$, with one fitted set of banks. The reconstruction arm supplies a real backbone code directly to G2, bypassing both the autoregressive draw and R1; G2 and the joint sweep then run as deployed. It tests those down-

TABLE II. The two-level model scored: mean (seed range, max – min) over three generation seeds at $n = 1000$. Both arms run their hot reads at the operating point of Sec. IV C; the reconstruction arm bypasses the draw and R1 with a real backbone code; the end-to-end arm draws from the label alone.

arm	critic	div	tone	std-ratio
real references	0.8860	0.4866	2.362	1.00
reconstruction	0.8963 (0.0110)	0.5184 (0.0015)	2.004	1.13
end-to-end	0.9007 (0.0310)	0.5487 (0.0052)	1.590	1.14

stream stages together, rather than isolating the draw alone.

The reconstruction and end-to-end critic scores are 0.8963 and 0.9007, against 0.8860 for the first 100 validation images per class. These are label-agreement scores, not evidence that generated images match or exceed real-image quality. A classifier can score class prototypes highly. End-to-end tone variation is lower than real (1.590 against 2.362), while contrast is higher (1.14 against 1.00) and exceeds the stated 1.10 alarm. The seed ranges are descriptive; they are not confidence intervals or a test of equivalence.

The generated batch shows no memorization signature: over 5120 drawn images every joint and backbone code is distinct, none is an exact copy of a training image, and their nearest-neighbour distances to the training set sit at or beyond those of held-out real images (L_2 first percentile 1.86 against 1.47, median 3.70 against 3.43). The code still changes after one joint sweep: the second sweep changes 82 of 226 addresses per image, the third 44, and the fourth 26.

C. The comparator sets the draw temperature

Every hot read in this paper runs at a modeled read voltage of $V = 10\text{ mV}$, with the pulse width settling-limited, $t_b = \max(2\text{ ns}, 5RC)$. We assume a minimum read voltage of 10 mV, called the sense floor. At this voltage, the target 1 mV trimmed comparator offset is 10% of full scale (Sec. II A). Kickback is a separate, unresolved readout requirement (Appendix B). At this voltage, the model gives device read noise of 0.02–0.04 across the backbone chain. Flicker accounts for 70% of the variance on the first read, when only the replicated label is present. Its share falls to 5% by the last read, where thermal noise dominates. The model’s thermal noise stays nearly constant along the chain. Settling time cancels the conductance dependence, leaving the kT/C term of Sec. II A with node capacitance fixed at 100 fF. The geometry of Appendix B puts the node at about 12 fF on the first read and 161–278 fF thereafter, so by that geometry the first read is noisier than modeled, about 0.07, and the later reads are as modeled. At this voltage, the lanes supply less than 40% of the required draw noise on the first read and less than 10% thereafter. Their thermal

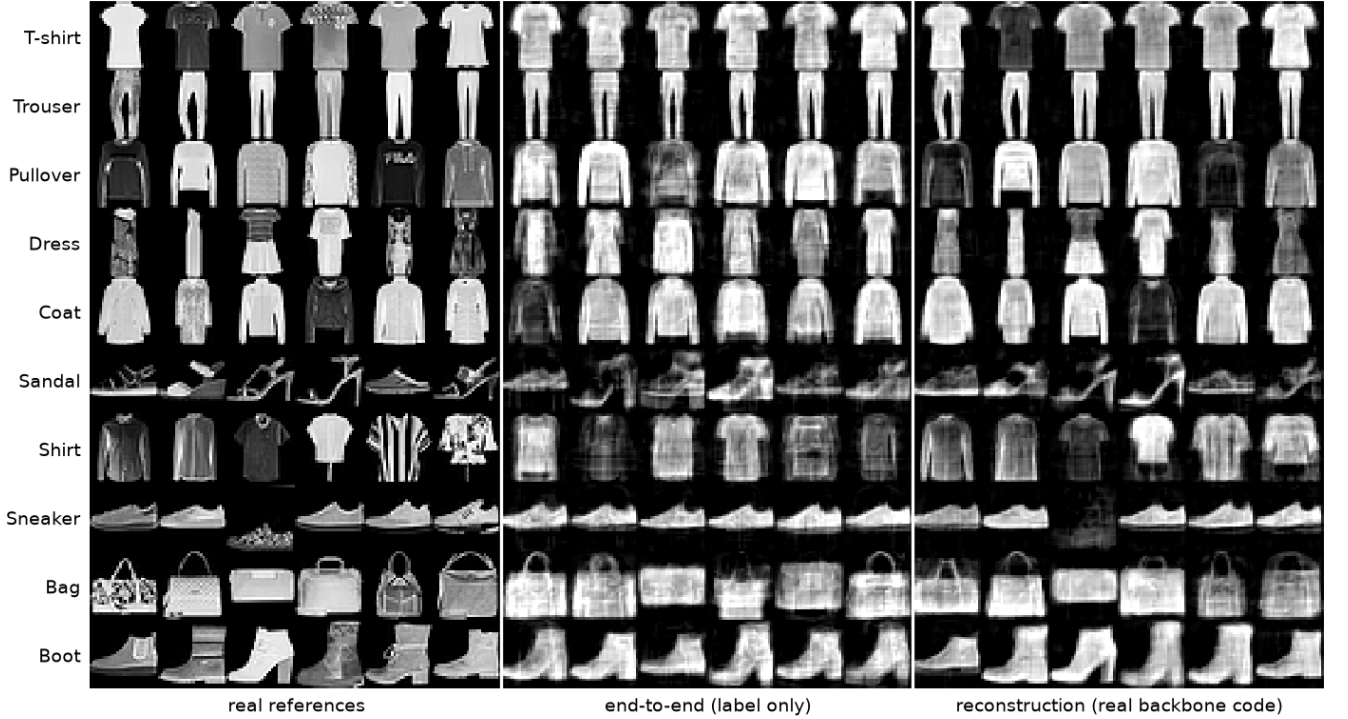


FIG. 3. **Given a real backbone code, the patch level and joint sweep produce images with similar classifier agreement to real ones.** Real references (left), end-to-end draws from the label alone at the operating point of Sec. IV C (center), and the reconstruction arm (right), six per class. The strips are the first six samples per class from reporting seed 960 ($n = 1000$), with no sample selection.

TABLE III. Comparator design classes and the numbers this paper models against: kT/C plus preamplifier thermal noise, referred to the input; offset from transistor mismatch [11]. The last column refers the offset to the 10 mV hot read, where a lane’s output spans ± 1 . A digital trim removes offset and leaves the noise of the underlying class.

class	noise (rms)	offset	offset / 10 mV
raw dynamic latch	0.5–2 mV	5–20 mV	0.5–2
auto-zeroed / chopped	50–200 μ V	0.1–1 mV	0.01–0.1
either, trimmed	as above	tens of μ V	~ 0.003

contribution is set by the node’s kT/C noise. Supplying enough noise from the node alone would require a read voltage near 1 mV. Circuit settings can increase the comparator term v_n/V in Eq. (4), which is independent of y and m (Table III).

The sweep. We swept v_n/V at $V = 10$ mV from 0 (lanes only) to 0.20 on the grayscale end-to-end draw and to 0.50 on the binary-trained two-level draw of Sec. V C, holding every other component as deployed. Both stacks improve with added comparator noise (Fig. 4): the best grayscale critic occurs at the tested endpoint $v_n/V = 0.20$, and binary FID is lowest at 0.2–0.3. The binary optimum lies on two grid points added after a first pass. A 7.5 mV point tested before the register was introduced

exceeds its ceiling and is not reported.

The comparator-noise level is a tunable model parameter. The emulator sets comparator noise with an 8-bit trim register: $v_n = 100 \mu\text{V} + \text{code} \times 20 \mu\text{V}$ for codes 0–255. We report noise levels by code. The full test grid uses codes 3, 20, 45, 95, 145, and 245. The grayscale critic sweep uses the first four; the grayscale FID sweep uses the first five plus a device-only point (Sec. V C). No comparator has zero input-referred noise, so the register has no zero-noise setting. We use $v_n/V = 0$ only to model device noise alone. At $V = 10$ mV the register’s ceiling of 5.20 mV caps v_n/V at 0.52, which is where the sweep stops.

At the optima the comparator supplies 97–99% of the read’s variance in the grayscale draw and 99–100% in the binary one as modeled; at the first read’s node capacitance of Appendix B the grayscale share on that read is about 90%. At fixed $v_n/V = 0.10$, raising the read voltage from 10 to 50 mV reduces lane thermal noise five-fold with little change in the metrics: grayscale critic 0.8837 versus 0.8840 and binary FID 10.73 versus 10.99. This supports comparator noise as the main control of sampling temperature. The two terms also differ along the chain. Device noise falls as $1/\sqrt{m}$, making it greatest early in the chain, when the pool is small, and least at the end. Comparator noise stays constant along the chain.

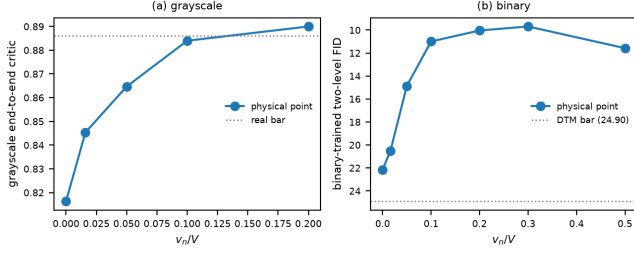


FIG. 4. **The comparator’s noise setting is the draw temperature.** (a) The grayscale end-to-end critic against the comparator’s input-referred noise relative to the read voltage, at $V = 10$ mV, with the real bar drawn as a line. (b) The same sweep on the binary-trained two-level stack, scored as FID under the protocol of Sec. VC, with the DTM reference arm’s FID drawn as a line.

The design requirement. No comparator class reaches these operating points on its own. At a 10 mV read, a dynamic latch’s own kT/C and preamplifier noise can reach $v_n/V \approx 0.2$ with low preamplifier bias. This reaches the grayscale optimum of 0.20. The binary optimum of 0.2–0.3 may require still lower bias or an added input noise source. But the same latch carries 5–20 mV of offset, which at a 10 mV read is 0.5–2 of the full range of a lane’s output (Table III), and no shift of the weights cancels an offset larger than the range. Static offset and read noise require separate control mechanisms. CHILL ZEN requires comparator offset below about 10% of the read voltage at every operating bias: 1 mV at a 10 mV read. Each lane needs a one-time trim if its offset exceeds this limit. Input-referred noise must be adjustable from near zero to about 20–30% of the read voltage, using reduced preamplifier bias or an added noise source. A higher hot-read voltage allows more offset if v_n/V stays fixed, as the 50 mV test above supports. At 50 mV, the allowed offset increases fivefold, but a hot read at $v_n/V = 0.20$ needs 10 mV rms of comparator noise.

The cold reads. The deployed generation runs its two cold sweeps at zero read noise. A physical cold read has a floor: at 50 mV the combined noise comes to about 0.003 of the read. To measure what that floor does, we ran both sweeps at zero noise, at the floor, and at ten times the floor, on the same hot draws, three seeds at $n = 1000$. The noise flips decisions, 9 of the 128 R1 addresses and 12 of the 226 joint addresses per image at the floor, and a quarter of the R1 addresses and 52 of the joint addresses at ten times it. The metrics do not move: critic 0.890 against 0.890 at the floor and 0.893 ± 0.024 against 0.890 ± 0.012 at ten times it, with diversity, tone, and contrast unchanged. The aggregate metrics show no detectable effect of replacing noiseless cold reads ($T = 0$) with the physical noise floor. We did not repeat the comparison in Sec. VC with that floor applied. Appendix B gives conditional noise-energy estimates; the bandwidth, noise, and energy have not been demonstrated together in one comparator.

D. What the controls establish

The reconstruction arm feeds real backbone codes to G2 and the joint sweep, and the critic holds. This test checks those two banks but does not locate the source of the final tone and contrast errors. It skips the draw and R1, and its classifier score does not fully describe the image distribution. The reconstruction FIDs of Sec. VC are likewise a reference for the code, not a bound on a generator.

At the operating point in Sec. IV C, comparator noise is fixed at $v_n/V = 0.20$ for every draw read. Contrast nearly matches the reconstruction arm (std-ratio 1.14 versus 1.13), while tone spread is lower (1.59 versus 2.00; Table II).

E. Design studies

Two scaling studies set the deployed code, both measured on the pipeline this paper describes.

(a) *Capacity and alphabet scaling.* At fixed alphabet size and keep probability $p = 0.5$, increasing from 64 to 128 books raised the backbone-only critic score from 0.813 to 0.834. The larger-alphabet configuration scored lower: 0.7875 for 64×32 at 320 bits versus 0.8344 for 128×16 at 512 bits. Both alphabet size and bit count changed in this comparison. A global code removes tile seams by construction: the mean pixel step across tile boundaries, relative to elsewhere, is 0.93–1.07 for the global code with real images at 1.100 (Fig. 5).

(b) *Codec rate-distortion.* At 512 bits, the global dropout-AQ code achieves its lowest relMSE, 0.0500, at keep probability $p = 0.75$. The deployed $p = 0.5$ code has relMSE 0.0612 but greater cross-address coherence, as described below. The two-level residual code reaches relMSE 0.0382 (Fig. 6). As plain additive quantization ($p = 1$) scales up, unused atoms and stalled search reduce its accuracy. The lowest reconstruction error occurs at $p = 0.75$. Using $p = 0.5$ raises that error by 12–22% but provides 2.2–3.6 times the cross-address coherence needed for generation. Both increases are largest at 128 books.

V. COMPARISON WITH THE DTM/DTCA

A. The design points

The two systems differ as design points (Table IV); our reading of the DTM/DTCA follows Ref. [5] and its released software [22].

The DTM chain keeps each step easy to mix, but its sampling cost still depends on the number of steps and iterations. CHILL ZEN has no mixing phase, so we report no mixing time. The remaining challenge is the quality of its draws. Both systems obtain their randomness from a circuit at each node, and what separates

TABLE IV. Two thermodynamic design points. DTM entries follow Ref. [5]; latency is derived from its stated timings.

	DTM / DTCA [5]	CHILL ZEN (this work)
generative principle	reverse a noising chain; each step an EBM sampled by Gibbs	draw an address code hot; one cold repair sweep reconciles
sampling procedure	$K_{\text{mix}} = 250$ Gibbs iterations per step, $T = 2-8$ steps	128 autoregressive reads + 3 parallel sweeps $\approx$ 131 sequential steps
single-image latency (conditional)	$2TK_{\text{mix}}\tau_0 \approx 400 \mu\text{s}$ at $T = 8$	131 sequential steps plus one parallel decode read; about $8 \mu\text{s}$ if each complete step takes 62.5 ns
random draws per image	$TKL^2 \approx 8 \times 250 \times 4900 \approx 10^7$ conditional updates	about 3600 hot lane reads
randomness source	dedicated subthreshold-transistor RNG, free-running, $\tau_0 \approx 100 \text{ ns}$	one comparator per lane, its input-referred noise settable; the pooled weights' own read noise is the minor term
temperature control	τ_0 fixed by circuit design	the read voltage and the comparator's noise setting; cold reads are modeled as noiseless
noise character	bias-programmable Bernoulli source	a flat settable term from the comparator, plus a state-dependent device term that pooling quenches
weights and sampler	separate circuit elements (couplings; RNG nodes)	the stored weight, read through one comparator per lane that is also the sampler
data representation	binarized Fashion-MNIST, ± 1 spins	grayscale Fashion-MNIST, 904-bit address code; binarized host-side for Sec. VC
rendered output	binary spins; grayscale via a multi-spin embedding; color via a GAN-tuned neural decoder	float grayscale via a fixed linear decode
learning	Monte Carlo gradient estimates from equilibrium samples (host optimizer)	local lane instructions for banks; host-fitted codebooks

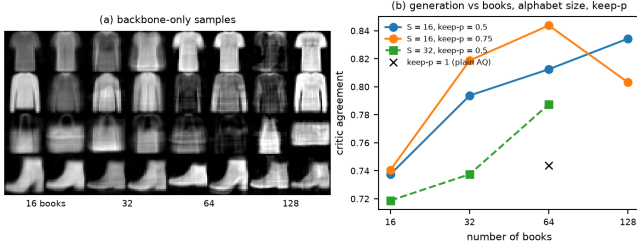


FIG. 5. **A single additive code over the whole canvas has no patch seams and improves with the number of books at fixed alphabet.** (a) Backbone-only samples at 16, 32, 64, and 128 books. (b) Critic agreement against the number of books and the alphabet size: the larger-alphabet configuration scored worse, but its lower bit count prevents attributing the difference to alphabet size alone. Plain additive quantization ($p = 1$) underperforms the dropout code.

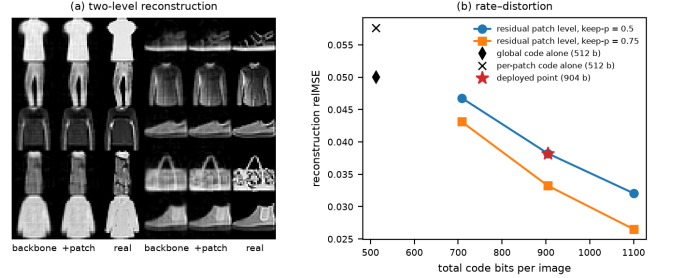


FIG. 6. **The residual construction reaches the program's lowest reconstruction floor.** (a) Backbone decode, backbone plus patch decode, and the real image, one example per class. (b) Rate-distortion of the residual patch level over the 512-bit global backbone; the deployed 904-bit point reaches relMSE 0.0382, and the global code alone outcores a per-patch code at rate parity.

B. Energy

For a read whose noise is thermally limited, the referred noise and the dissipation are

$$\sigma^2 = \frac{2k_B T}{tV^2 G_\Sigma}, \quad E = V^2 G_\Sigma t, \quad (9)$$

where G_Σ is the selected-pool conductance and t the read pulse width, and their product eliminates the device en-

them is how many draws an image needs, about 3600 hot lane reads against about 10^7 conditional updates at their stated $K_{\text{mix}} = 250$. The systems do different amounts of work per step, so step counts alone cannot compare their energy use. We therefore estimate energy in joules; neither system has been fabricated at the target scale.

tirely:

$$E_{\text{read}} \sigma^2 = 2k_B T. \quad (10)$$

The energy of one stochastic sample at noise level σ is set by thermodynamics alone, for a pooled lane read exactly as for a single device. For a comparator that samples the node at settling-limited timing the realized product is $5k_B T$ (Appendix B). The identity covers only the lane’s noise contribution, with an energy cost of roughly 10^3 – $10^4 k_B T$ per read at the deployed noise levels. The lane supplies just 1–3% of the draw’s variance at the operating point in Sec. IV C. The comparator supplies the remaining noise at about 10 fJ per lane read, or roughly $10^6 k_B T$. A DTCA draw costs less: its RNG uses a measured 350 aJ per bit, and a full cell update uses about 2 fJ, including the RNG, bias, clock, and communication [5]. The identity sets the energy of only the smaller noise contribution and is not used in the estimates below. Total energy also depends on how many draws each system needs: the accounting below counts about 3600 noisy reads per image, where the chain counts $TKL^2 \approx 10^7$ conditional updates.

The per-image energy is a conditional estimate from the stated geometry of Appendix B: the neural lane of Ref. [14], one 4×4 unit-crossbar pair per address space on a 50 nm pitch, the address of a read broadcast on select lines across every lane of a bank, and the unselected lines floating (Limitation 2). The model counts three energy terms per image: address delivery at the select supply, E_{sel} ; divider reads at $5C_{\text{node}}V^2$ each, independent of device conductance, E_{read} ; and one comparator and lane buffer per lane read, E_{readout} . Readout assumes a continuous-time comparator; Table XI gives alternatives. Rail energy is below 10^{-11} J and is omitted. It is applied to the exact operation count of the deployed system, 10 064 lane reads and 1.78×10^6 address assertions over five banks totalling about 2.9×10^7 synapse pairs, with constants that are each a stated dimension or a textbook value (Table XIII). The decoder is the fixed linear map of Eq. (7): with the codebook atoms stored as differential pairs, one output lane per pixel, and every code address present at the end of a generation, the pooled read of Eq. (5) returns the sum of the selected atoms at a fixed scale. Decoding accounts for about 8% of all reads. This paper’s emulation runs that map in host floating point, as it does the render and the judging; the model counts it as the 784 cold lane reads it is designed to be. Running it on lanes, with the atoms quantized to a pair’s resolution and read at 50 mV, has not been emulated and its precision is not measured. The DTM has no decode for binarized Fashion-MNIST because its spins are the pixels. This saving applies to binary pixels. The DTM uses a multi-spin embedding for grayscale and a separately trained neural decoder for color (Table IV), both excluded from its energy estimate. Our Sec. IV system generates grayscale images, which we binarize for the comparison. The model’s address-delivery term is what Ref. [5] calls E_{comm} and reports as about half of its

TABLE V. Modeled energy per image of the deployed two-level system at the nominal point of the model and at its stated sensitivities, against the 8-step DTM chain’s 1.57×10^{-8} J [5]. Appendix B gives the terms.

operating point	J/image	ratio
nominal: 4×4 , 0.8 V select	2.0×10^{-9}	7.9× less
low-swing select, 0.5 V	8.8×10^{-10}	17.7× less
7 nm-class periphery	8.2×10^{-10}	19.0× less
7 nm-class periphery, 0.5 V	4.2×10^{-10}	37.0× less
pessimistic corner ^a	8.1×10^{-9}	1.9× less

^a 16×16 arrays, separate *a*- and *b*-side select lines, 1.0 V.

budget.

Address delivery is 91% of the total at the nominal point, readout 8%, and the divider reads 1%. Table V reports the nominal energy estimate and its sensitivity to the select supply and peripheral circuit design rules. The two 7 nm-class rows show how an advanced process node affects this design. Ref. [5] uses an advanced node but does not identify it. For that nominal geometry and those circuit assumptions, the estimate is about 2×10^{-9} J per image, eight times below the DTM chain, with the non-7 nm sensitivity rows running from about two to eighteen times below. These are scenario calculations, not uncertainty bounds on a working implementation. The floating-line nominal crossbar does not implement the isolated reads used for the quality results. Table XII separately puts a selector-column geometry without that sneak path at 1.6 nJ, using the same assumed readout and decoder energies. Thus the modeled advantage persists in an array geometry consistent with isolated reads; the peripheral assumptions still apply. The chain’s 1.57×10^{-8} J is read from their Fig. 1; their text states about $1.6 T$ nJ, which at $T = 8$ is 1.28×10^{-8} J and a nominal ratio of 6.4 (Appendix B). The single-knob sensitivities of Appendix B move the ratio between 4.6 and 7.9. Because the model depends on counts only, the binary-trained arms of Sec. V C, whose banks and operation count are those of their grayscale twins, carry the same energies.

Time per image. A full Gibbs iteration in the DTCA takes about $2\tau_0$, with $\tau_0 \approx 100$ ns [5]; at $K_{\text{mix}} = 250$, the eight-step chain requires about 400 μ s of sampling time. The DTCA waits on its RNG, whereas CHILL ZEN could be limited by settling and readout: Appendix B’s illustrative 500 MHz comparator bandwidth corresponds to a noise correlation scale of about 0.3 ns. CHILL ZEN uses 131 sequential generation steps plus one parallel read for the proposed lane decoder. If each complete step fits the 62.5 ns window used to budget readout energy, an image takes about 8 μ s, roughly fifty times faster than the chain. Adopting Appendix B’s assumed 30 μ s image time gives a ratio of about thirteen; that leakage example establishes no timing bound. The readout window does not establish a complete instruction cycle, including address delivery, settling, and commitment; the comparator and decoder limitations already stated ap-

TABLE VI. The evaluation protocol used here, from the replication code [24] linked from Ref. [22].

element	value
extractor	InceptionV3, 2048-d pool features
resize channels	256×256 , bilinear, antialiased grayscale tiled to 3, then scaled to $[-1, 1]$
distance	Fréchet distance via matrix square root
generated n reference	5120, i.e. 512 per class shipped μ and Σ over the 60 000-image train split
binarization	threshold 0.1 on the $[0, 1]$ image

ply. These estimates exclude initialization and external transfer. With separate hardware for all eight EBMs, an ideal DTM pipeline could deliver one image every $50 \mu\text{s}$ after filling, while retaining about $400 \mu\text{s}$ latency.

C. Binarized Fashion-MNIST and the published DTM reference

Reference [5] reports quality as FID [23] on binarized Fashion-MNIST. Comparing against it means binarizing our renders and scoring them the same way, which requires knowing what the same way is. Their paper states the metric and nothing else: no sample size, no feature extractor, no binarization threshold, and their released library [22] contains no FID code. The library links to a DTM replication repository [24] containing evaluation code and precomputed reference statistics for reproducible scoring (Table VI). Its logged eight-step runs score FID 25–27, and its author credits a DTM author for guidance.

Two elements are unconventional: images are resized to 256×256 rather than 299, and binarized at 0.1 rather than at the midpoint. The replication’s source specifies a binarization threshold of 0.1. We checked this against its reference statistics: over 2000 training images, the mean feature vector at 0.1 had one eighth the relative L_2 error of the next candidate. Using the replication’s unmodified feature extractor and distance code, all 60 000 training images binarized at 0.1 scored FID 0.000039 against the supplied μ and Σ . Agreement at that level across a 2048×2048 covariance fixes every element of the protocol at once. It establishes that our scoring path is identical to the replication’s. The canonical test split scores 2.142 at $n = 5120$. This is a finite-sample real-data reference, not a universal FID floor or a held-out test of CHILL ZEN, whose fitting partition includes most canonical test images. The number of samples used for the published DTM scores is unknown.

The grayscale-trained rows use the Sec. IV systems with weights unchanged. They generate grayscale values in $[0, 1]$; the host only applies the threshold. Compara-

TABLE VII. Binarized Fashion-MNIST under the protocol of Table VI: the replication evaluation code, reference statistics, threshold, and sample count. DTM rows are published values, not reruns through this pipeline. Reconstruction rows use real codes from our 68 000-image fitting partition, which differs from the canonical reference split. Diversity is the within-class mean pairwise pixel disagreement of the scored images. Energies are the nominal point of the model of Sec. VB; DTM energies are from Fig. 1 of Ref. [5]. Each grayscale row is drawn at its best level on the register grid of Sec. IV C, at $V = 10 \text{ mV}$: the two-level stack at $v_n/V = 0.10$, the 128- and 64-book one-level systems at 0.05, the 32-book system at 0.016, the 16-book system at 0.10, and the binary-trained rows at 0.30. Every level’s score is recorded in the companion repository.

system	FID	div	J/image
real fitting images (control)	1.91	0.181	—
128-book reconstruction, grayscale	14.21	0.163	—
128-book reconstruction, binary	9.57	0.167	—
binary two-level reconstruction	2.83	0.176	—
CHILL ZEN two-level	18.09	0.156	2.0×10^{-9}
CHILL ZEN one-level, 128 books	22.11	0.152	7.5×10^{-10}
CHILL ZEN one-level, 64 books	22.62	0.143	2.5×10^{-10}
CHILL ZEN one-level, 32 books	25.76	0.120	9.9×10^{-11}
CHILL ZEN one-level, 16 books	44.15	0.116	4.9×10^{-11}
binary-trained two-level	9.69	0.147	2.0×10^{-9}
binary-trained one-level, 128 books	15.46	0.130	7.5×10^{-10}
binary-trained one-level, 64 books	15.10	0.129	2.5×10^{-10}
DTM, 8 steps [5]	24.90	—	1.57×10^{-8}
DTM, 2 steps [5]	32.60	—	3.92×10^{-9}

tor noise levels were selected for the best reported FID. The binary-trained rows exist because that path carries a training-information asymmetry: an 8-bit training signal thresholded at scoring time is richer than the 1-bit data the DTM trains on. They are the same architecture, codec objective, and teaching recipe, fit end to end on the 0.1-binarized data, so their renders are $\{0, 1\}$ images and the threshold is the identity on them.

Under the stated energy models, the two-level system uses 2.0 nJ per image, about one eighth of the published eight-step DTM’s 15.7 nJ. It scores FID 18.09 versus the DTM’s reported 24.90. Direct binary training lowers our FID to 9.69 at the same modeled energy. This compares emulated generation quality and estimated hardware energy with the published DTM results. The energy ratio follows from the two accountings; it is not a measurement of either complete machine. Our scoring path is verified against the replication, and the quality comparison retains the protocol and training-split qualifications stated above.

The grayscale 128-book reconstruction reference scores 14.21. It uses only the backbone, so it is not a reference for reconstruction through the full two-level codec. Nor does reconstruction FID bound generation FID: FID compares feature distributions and does not decompose into additive codec and draw errors. End-to-end pixel disagreement is 86% of the real fitting-image reference; that statistic alone does not establish mode coverage or

locate the cause of its shortfall. The one-level frontier was swept on the same six-level grid, and each arm has its own optimum: the 128- and 64-book systems at $v_n/V = 0.05$, the 32-book system at 0.016, the 16-book system at 0.10. At those levels the 128- and 64-book systems clear the DTM bar (22.11 and 22.62 against 24.90) and the 32- and 16-book systems do not (25.76 and 44.15). The critic’s best level and the FID’s best level differ for the two-level stack too: the critic of Sec. IV B peaks at $v_n/V = 0.20$, where the same renders score FID 21.95 under this protocol, against 18.09 at 0.10. Table II reports the critic’s optimum and this table the FID’s; both are levels of one declared grid, and both metrics at every level are recorded.

The binary-trained arms. Training on binary images gives lower FID under this protocol. The binary-trained two-level stack scores FID 9.69, against 18.09 for the grayscale-and-threshold path and 24.90 for the DTM, while the full binary two-level reconstruction reference scores 2.83. The 9.57 row is the backbone-only reconstruction. Neither reference is a lower bound on generator FID. Thus binary fitting works better for this architecture under the chosen protocol; this does not remove the training-split asymmetry with the DTM. Every binary-trained configuration below $v_n/V = 0.20$ fails the diversity screen. This supports the finding in Sec. IV C: device noise alone is insufficient for sampling, so the comparator must add noise.

Comparison with their baselines. The DTM rows of Table VII are transcribed from the published paper; their scoring path has not been independently verified. We extracted the GAN, VAE, MEBM, and DDPM values in Fig. 8a from the vector source of the DTM paper’s Fig. 1 and checked the VAE values against its Appendix F. The DTM authors trained and scored these baselines; we cannot inspect the full procedures. The published best values are 24.9 for DTM, 33.5 for MEBM, 26.3 for GAN, and 17.9 for VAE. The best DDPM point decoded from the vector figure is 12.2, below the plotted axis range; the best visible point is 16.4. These provide context for CHILL ZEN’s 18.09 and 9.69, but unverified evaluation details prevent a controlled ranking or a claim of parity with the DDPM.

Three audits. First, every CHILL ZEN row is drawn with the label given, 512 per class. The replication reports both scoring modes for 44 runs. In every run, the best clamped FID is lower than the best free FID, by a median of 3.0 points. The DTM paper does not specify which mode its Fig. 1 reports. Our free mode chooses a label uniformly, then runs the same draw. FID rises by less than half a point relative to clamped mode: 10.10 versus 9.69 for binary training and 18.27 versus 18.09 for grayscale training. Class balance is set by the label prior. Second, a simple independent-Bernoulli sampler scores poorly under the same protocol: FID 180.5 with class information and 258.1 without it. Third, an exact-copy and nearest-neighbor audit of the binary-trained renders against the actual 68 000-image fitting

partition finds all 5120 distinct and no training-image copies. The 2000 images excluded from fitting include five exact training-image duplicates; the nearest-training-neighbor Hamming medians are 25 bits for generated images and 23 for that reference. These checks found no exact-copy signature; they do not rule out all forms of memorization. The reference images were used for model selection.

The threshold moves FID more than the model does. The evaluation protocol sets the binarization threshold. Changing it shifts FID by more than the gap between the compared systems. Scored against their reference, the same 60 000 real train images binarized at 0.5 rather than 0.1 give FID 29.75, worse than their 8-step DTM. Figure 7 shows why: at 0.5 the smooth mid-tones of a render straddle the threshold and break into speckle, while at 0.1 they resolve into clean silhouettes. A number scored under an unstated data convention is not comparable to anything, ours included.

VI. LIMITATIONS

1. All results are emulator results with bit-exact lane semantics. Energy numbers are modeled from a stated geometry, not measured, and are reported with their sensitivities for that reason. Sending read addresses to the arrays at logic voltage dominates energy use. This cost depends on the peripheral transistor pitch and select supply. In the pessimistic scenario, the DTM’s estimated energy is about twice CHILL ZEN’s. These scenarios do not bound omitted circuit energy. The model covers address delivery, read, and readout energy per image and not die area, leakage beyond the bound of Appendix B, or the one-time energy of programming the roughly 5.7×10^7 memristors the banks require.
2. Without selectors, the selected symbol contributes only 44% of the nominal crossbar’s conductance during a read or feedback pulse. Its six row and column neighbors contribute 48% (Sec. B 5). Every result of this paper is at the emulator’s ideal fidelity, where a read returns the selected pair alone; the effect of the sneak share on teaching, address resolution, and write disturb has not been measured. The 16×1 selector column of Table XII has no sneak path and is the geometry we would build.
3. The comparator is a model; no circuit has been designed. Its noise is one flat term set per read, its offset is assumed trimmed per lane to below a tenth of the read voltage at each bias setting, and its energy is one estimated constant, with the readout circuit’s alternatives in Table XI. The operating points of Sec. IV C state what a comparator must supply; they do not show a circuit that supplies

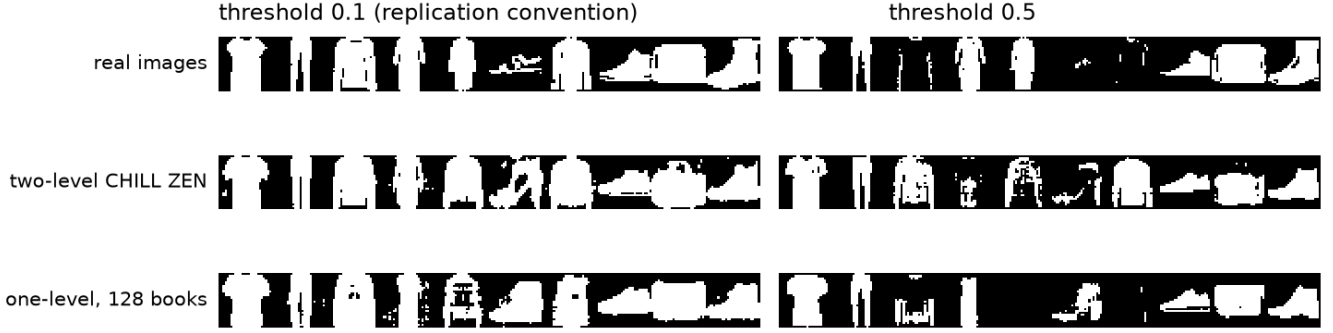


FIG. 7. **The binarization threshold changes every image, and it is not a model choice.** The same renders, one sample per class, thresholded at the replication’s 0.1 (left) and at the midpoint (right). Real Fashion-MNIST scored against their own reference gives FID 0.000039 at 0.1 and 29.75 at 0.5, so the convention alone spans more than the gap between any two systems in Table VII.

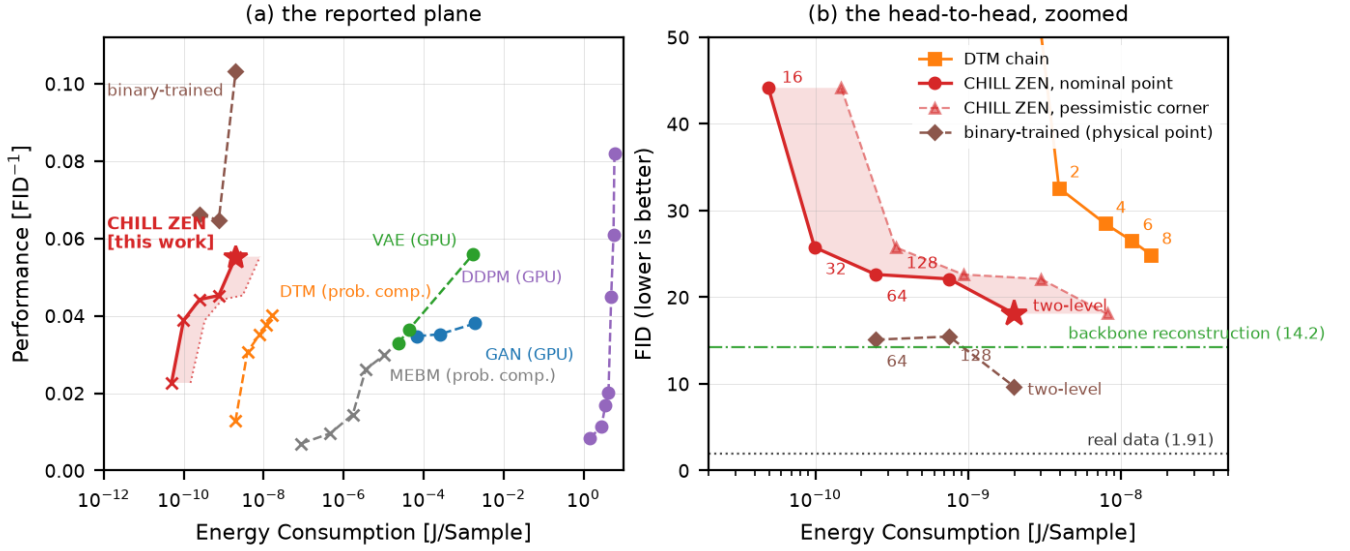


FIG. 8. **CHILL ZEN’s performance versus energy.** CHILL ZEN FIDs use the replication protocol of Table VI; the other series are published values whose scoring path we could not verify. The CHILL ZEN frontier spans one-level systems of 16, 32, 64, and 128 books and the two-level system (star) at the nominal point of the energy model; the shaded band extends each point to the model’s pessimistic corner (Appendix B). The binary-trained arms (diamonds) are plotted at their computed energies. (a) The axes of Fig. 1 of Ref. [5], with their five reported series re-plotted from their data. (b) The same frontier against the DTM chain alone, drawn as FID so the margin is readable, with the backbone reconstruction and real-data references. Each CHILL ZEN point is at its own best comparator level; the two-level system and the 128- and 64-book one-level systems clear the DTM chain, and the 32- and 16-book systems do not. The energy values for this work are estimates from the model of Appendix B.

it. The emulator’s device constants are placeholders pending measurement. The thermal term needs no assumed device constant: at the settling-limited pulse, it follows kT/C . The model averages noise over the pulse, while a strobed comparator samples it at one instant. Sampling raises the lane’s thermal noise by about a factor of 1.6 without changing the conclusions. Only the flicker term requires a device-specific noise constant. It contributes 5–70% of the lane’s noise variance; the lane itself con-

tributes only a few percent of the draw’s variance. Extending the model from its $1\mu s$ reference pulse to nanosecond pulses assumes the $1/f$ noise band reaches those frequencies.

4. The quality bar of Sec. IV is a pixel-critic agreement score, not FID, because Ref. [5] establishes no grayscale reference. The end-to-end critic score exceeds both the reconstruction score and the real-image reference by less than its variation across seeds. This measures label agreement. Beyond the

tests in Sec. IV D, we have not identified which features of the generated distribution affect the score. Contrast still exceeds the alarm threshold: std-ratio 1.14 versus 1.10.

5. Our systems are fit on a 68 000-image split of the full 70 000-image dataset, while the reference statistics of Sec. V C are over the 60 000-image train split that Ref. [5] trains on; the training sets are neither identical nor the same size.
6. The scores are validation numbers, not held-out test numbers: the evaluation partition was used for epoch and hyperparameter selection, and each row of Table VII is reported at the comparator level that gave its best FID. There is no gradient-trained baseline on this code and connectivity, so the paper does not claim the local rule outperforms gradient training. Conditioning is label-clamped; the unconditional mode of Sec. V C draws the class from the prior rather than learning it.

VII. CONCLUSION

CHILL ZEN combines a host-fitted additive code with locally taught lane banks and a fixed generation procedure. The emulator generates without waiting for an equilibrium distribution, using about 3600 noisy lane reads per image. Pooling leaves too little device noise at the assumed read voltage, so the comparator must supply most of the draw variance. On binarized Fashion-MNIST the two-level system scores FID 18.09, or 9.69 when trained directly on binary data, against the eight-step DTM’s published 24.90. The chain’s estimated energy is 7.9 times CHILL ZEN’s nominal estimate. The selector-column design also has a lower energy estimate than the chain and eliminates the nominal array’s sneak paths. These comparisons rest on the stated scoring protocol and circuit assumptions. The architecture saves energy mainly by reducing sampling work: it generates each code in a fixed sequence of reads with no equilibrium mixing phase.

Appendix A: List of acronyms

Appendix B: The energy model

The energy figures of Sec. V B come from the model below, a physical estimate from a stated geometry; the script that computes it is named in Sec. B 10.

1. Scope

The model counts three terms per generated image: the delivery of each read’s address to the memristor ar-

TABLE VIII. Acronyms used throughout this work.

Acronym	Meaning
AAT	Activation Address Tuple
AHaH	Anti-Hebbian and Hebbian
AQ	Additive Quantization
AR	Autoregressive
CHILL ZEN	Comparator-Heated In-memory Lane Learning with Zero-Ended Noise
DTCA	Denoising Thermodynamic Computer Architecture
DTM	Denoising Thermodynamic Model
EBM	Energy-Based Model
FID	Fréchet Inception Distance
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
MEBM	Monolithic Energy-Based Model
RNG	Random-Number Generator
VAE	Variational Autoencoder
WTA	Winner-Take-All

rays, the divider reads themselves, and the per-lane read-out. It excludes rail supplies and leakage, estimated in Sec. B 8, and comparator kickback, which remains unverified. The estimate excludes weight-programming energy and die area, with no bound on either. This follows Ref. [5], which omits its own weight programming and reports only per-sample energy. The basis of comparison is stated in Sec. B 9.

2. Geometry

The lane is the neural lane of Ref. [14], Chapter 4. Its element is the *unit crossbar*, a small memristor crossbar. Row and column multiplexers select one device between source and drain lines. With no selector transistor at each memristor, floating unselected rows and columns allow current through other devices. A synapse is a pair of unit crossbars, the *a*-side between rail V_a and the output node and the *b*-side between the node and rail V_b . A lane is K such pairs, one per address space, with every pair’s midpoint tied to one output line V_y . Selection is the AAT: one address per space, broadcast to every lane of a bank.

The array is an integrated crossbar with a 20 nm via at each crossing and the memristor in the via, on a 50 nm pitch. Each space of the deployed system has 16 symbols, so the nominal unit crossbar is the smallest that holds them, a 4×4 array $0.20 \mu\text{m}$ on a side with two 4:1 pass-transistor multiplexers and $2 + 2$ address bits. A 16×16 array is carried as a sensitivity.

Three consequences of that layout fix the capacitances (Fig. 9):

1. *The select lines of a space run across the bank.* Every lane of a bank reads the same context, so the four row-select and four column-select lines of space

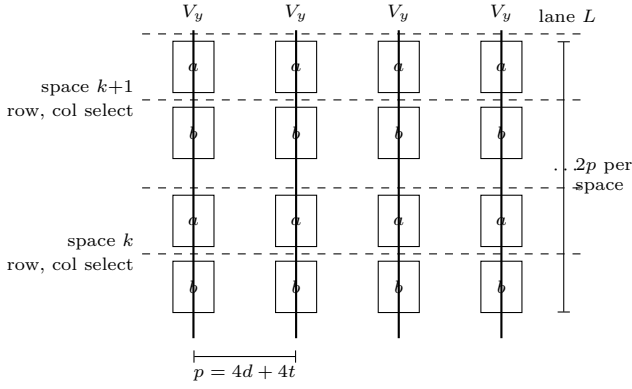


FIG. 9. The geometry the model assumes. Each box is one unit crossbar; a pair (a over b) is one synapse, and the pairs of a lane stack along its output line V_y . The dashed lines are the row- and column-select lines of a space, shared by every lane of the bank. A read charges the two select lines of every asserted space across all L lanes (term E_{sel}) and settles each lane's V_y node through the selected devices (term E_{read}). The lane pitch p along a select line is the 4×4 array plus one multiplexer stack of four pass transistors; the node is $2p$ long per space.

TABLE IX. Derived dimensions and capacitances, from the constants of Table XIII.

quantity	nominal	7 nm-class
transistor pitch t	$0.20 \mu\text{m}$	$0.06 \mu\text{m}$
lane pitch $p = 4d + 4t$	$1.00 \mu\text{m}$	$0.44 \mu\text{m}$
$C_{\text{seg}} = 2C_g + p c_w$ per lane	0.80 fF	0.29 fF
C_{node} , backbone and patch lanes ($K = 134$)	161 fF	131 fF
C_{node} , joint and decode lanes ($K = 232$)	278 fF	226 fF

k cross all L lanes. The a - and b -side multiplexers of a pair take the same address, so one row line and one column line serve both, each driving two pass gates per lane. Asserting a symbol charges two lines per (space, lane).

2. *The multiplexers set the lane pitch, not the array.* Along a select line one lane is one unit crossbar wide: the $0.20 \mu\text{m}$ array plus one stack of four pass transistors at the contacted transistor pitch t , so $p = 4d + 4t$ with d the crossbar pitch. At $t = 0.20 \mu\text{m}$ (a 28–40 nm-class periphery) $p = 1.00 \mu\text{m}$; at $t = 0.06 \mu\text{m}$ (a 7 nm-class periphery) $p = 0.44 \mu\text{m}$. Total energy is only weakly sensitive to via and crossbar pitch: doubling d to 100 nm changes it by less than 5%.
3. *The V_y node is $2K$ lane pitches long and carries $8K$ multiplexer drains.* The a and b arrays of a pair and the K spaces all stack along V_y . Its capacitance is $C_{\text{node}} = K(2p c_w + 8C_d)$ with c_w the wire capacitance per length and C_d the drain capacitance of an off pass transistor.

3. Operation count

The operation count of the deployed two-level system is exact, taken from the banks' lane and space counts. The generation of one image reads five banks (Table X). Two counts enter the model per bank: the number of (space, lane) assertions, LK , which is what the select lines must deliver, and the number of lane reads, which is what the nodes and comparators must resolve. The backbone draw and patch draw are hot reads at 10 mV; the sweeps and the decode are cold reads at 50 mV.

The backbone draw is 128 sequential steps, each reading a fresh group of 16 lanes with one more space asserted. Select lines span the bank, so selecting space k for group k also activates it in groups that have already read. Only 142 336 of the g bank's 276 576 select assertions contribute to a read. Segmenting the lines per group would save 48% of the g bank's E_{sel} , 7.5% of the total E_{sel} , with a driver per group per space; the model does not include it.

4. Energy terms

Per image,

$$E = E_{\text{sel}} + E_{\text{read}} + E_{\text{readout}} + E_{\text{rail}}. \quad (\text{B1})$$

Address delivery. For each bank, each space asserted per image charges one row line and one column line across every lane of the bank. A line charged once and discharged once per image dissipates CV^2 :

$$E_{\text{sel}} = \sum_{\text{banks}} LK \cdot 2 C_{\text{seg}} V_{\text{sel}}^2, \quad C_{\text{seg}} = 2C_g + p c_w. \quad (\text{B2})$$

This is the term Ref. [5] calls E_{comm} , the charging of the parasitic capacitance of the wires that carry a cell's state to its neighbors, and reports as about half of its budget.

The divider read. A lane read settles its V_y node through the selected devices and pass gates. With the pulse width set by settling, $t = 5RC = 5C_{\text{node}}/G_{\Sigma}$,

$$E_{\text{read}} = \sum_{\text{reads}} V_{\text{read}}^2 G_{\Sigma} t = \sum_{\text{reads}} 5 C_{\text{node}} V_{\text{read}}^2. \quad (\text{B3})$$

The pooled conductance cancels; the factor 5 is for rails at $\pm V$ about the node and a balanced pool, so that each side settles through V . Changes that affect only conductance alter settling time but leave read energy unchanged. These include device conductance range, size, write compliance, and sneak conductance in a crossbar with floating lines. Sneak conductance does change what the node reads: in a 4×4 array with floating unselected lines the selected device is 7/16 of the conductance the node sees. Sec. B5 derives that and the energy of each way out, and Limitation 2 states what follows for the results.

Equation (B3) also qualifies the thermodynamic identity of Sec. VB, Eq. (10). That identity, $E \sigma^2 = 2k_B T$,

TABLE X. The per-image operation count and each bank’s contribution at the nominal point. Lanes: g bank $(1 + 128) \times 16$, r bank 128×16 , patch 98×16 , joint 226×16 , decode 784.

bank	L	K	LK	reads	V_{read}	E_{sel} (J)
backbone draw (g)	2064	134	276 576	2048	10 mV	2.8×10^{-10}
backbone sweep (r)	2048	134	274 432	2048	50 mV	2.8×10^{-10}
patch draw	1568	134	210 112	1568	10 mV	2.2×10^{-10}
joint sweep	3616	232	838 912	3616	50 mV	8.6×10^{-10}
decode	784	232	181 888	784	50 mV	1.9×10^{-10}
total			1 781 920	10 064		1.8×10^{-9}

assumes the read integrates the node over the full pulse. A comparator that samples the node sees its $k_B T/C$ noise, $\sigma^2 = k_B T/(C_{\text{node}} V^2)$, and with $E = 5C_{\text{node}} V^2$ the realized product is $5k_B T$. The identity remains a floor on the lane’s own contribution to a hot read’s noise; it does not enter the accounting.

Readout. One comparator per lane resolves the read against the group’s shared falling reference (Sec. II C). The model assumes a continuous-time comparator with a preamplifier and latch drawing 200 nA from 0.8 V for 62.5 ns. This gives 10 fJ per lane read for bias current; the model assigns no separate energy to the decision. The lane buffer of the readout circuit of Ref. [14], “The Rank-Cut recoder in hardware,” a follower biased at 100 nA for the same window, dissipates 5 fJ:

$$E_{\text{readout}} = N_{\text{reads}} \times 15 \text{ fJ}. \quad (\text{B4})$$

The continuous-time comparator triggers when the reference crosses the lane voltage. This gives one noise sample per lane, as in the emulator, if the reference crosses the comparator’s $\pm 3\sigma$ band within one noise correlation time. The required slope is at least $6v_n \cdot 2\pi B$. At $v_n = 2 \text{ mV}$ and $B = 500 \text{ MHz}$, this is 38 mV/ns , enough to sweep the 20 mV range in about 0.5 ns. Reducing preamplifier bias increases noise (Sec. IV C): the input-referred thermal noise variance is $4k_B T \gamma B/g_m$, where $g_m = I/(nV_T)$. At fixed bandwidth-time product BT , the energy per read for rms noise v_n is $4k_B T \gamma nV_T V_{dd} (BT)/v_n^2$. With $\gamma = 1$, $nV_T = 35 \text{ mV}$, and an illustrative $BT = 3$, a 2 mV hot read uses 0.35 fJ. Lower noise requires more energy: 1.4 fJ at 1 mV (code 45), 15 fJ at 300 μV (code 10), and 139 fJ at 100 μV (code 0). These noise-energy examples use $BT = 3$, whereas 500 MHz over the model’s 62.5 ns window gives $BT = 31.25$ and energies 10.4 times larger. They therefore do not establish that the modeled comparator meets both the hot-read ramp condition and the cold-noise requirement. The cold-noise control measures grayscale sensitivity, not circuit feasibility or FID at that circuit’s noise level. Table XI compares the alternatives, printed by `experiments/38_readout_options.py`. A clocked comparator with a falling reference returns every lane’s rank, with one strobe and fresh noise sample per reference step. A clocked pairwise tournament returns only the winner, which is all generation uses, at one decision per lane. A continuous-time comparator returns both

winner and rank; its standing bias energy depends on the required noise level.

Rails. The rails V_a and V_b apply the instruction’s voltage during the read pulse and return to node potential between reads. Each lane therefore conducts only during its own read. Holding the rails active throughout generation would keep every selected device in the bank conducting. Charging the rails per read is bounded in Sec. B 8 and carried as zero.

5. Sneak paths and the ways out

The nominal 4×4 array has no selector and floats its unselected lines. With every device at a common conductance G the sneak network puts $1.29G$ in parallel with the selected device, so the selected device is 7/16 of what the node sees: a read is 44% the selected symbol, 48% its six row and column neighbors, and 8% the other nine, and a feedback pulse divides the same way. At a 16×16 array the selected share is 12%. The nodal solve is `experiments/36_sneak_paths.py`.

Table XII compares three ways out. Holding the node-side unselected lines at ground makes the read exact but shunts the node with $2.25G$ per side, a gain loss of about three that the read voltage must make up. Holding them at a replica of V_y is exact but needs a per-lane buffer whose bias the model cannot estimate. A 16×1 selector column uses one 1-of-16 multiplexer on the rail side and none on the node side. It eliminates sneak paths and charges one select line per assertion instead of two, using less energy than the 4×4 array. It requires twice as many select lines and 2.5 times as many leakage paths. Leakage remains negligible only because the rails pulse during each read (Sec. B 8).

6. Constants

Every constant is a stated dimension or a textbook value (Table XIII). The model has no fitted parameter.

TABLE XI. Five readout circuits on the deployed census with the nominal constants; every row is printed by `experiments/38_readout_options.py`. Clocked decisions dissipate 10 fJ each; continuous-time rows are bias times supply times the 62.5 ns window; every row carries the 5 fJ lane buffer. The noise column says how many independent comparator noise samples a lane’s decision passes through: the emulator computes one.

readout circuit	returns	noise samples per lane read	per lane read	readout share	ratio
continuous-time, 200 nA for 62.5 ns, as modeled	winner and rank	one, when the reference falls faster than the noise correlation time	15 fJ	8%	$7.9\times$ less
continuous-time, 1 μ A for 62.5 ns	winner and rank	one, at $\sqrt{5}$ lower noise	55 fJ	23%	$6.5\times$ less
clocked, binary search on the winner, 5 strobes	winner only	five, the first crossing decides	55 fJ	23%	$6.5\times$ less
clocked, full ramp, 16 strobes	winner and 4-bit rank	sixteen, the first crossing decides	165 fJ	47%	$4.5\times$ less
clocked tournament, 15 pairwise decisions per group	winner only	one per comparison, on lane differences	14 fJ	7%	$7.9\times$ less

TABLE XII. The energy of each way out of the sneak paths, by the model of Appendix B on the deployed census with the nominal constants; every row is printed by `experiments/36_sneak_paths.py`. The read column is for a 4×4 array with every device at a common conductance G . The two selector-column rows differ only in whether the sixteen pass gates sit beside the array or stack along V_y .

geometry	what the node reads	added circuitry	E_{sel} (J)	total (J)	ratio
4×4 , floating lines, as modeled	44% the selected symbol, 48% its six row and column neighbours, 8% the other nine	none	1.8×10^{-9}	2.0×10^{-9}	$7.9\times$ less
4×4 , node-side lines at fixed ground	the selected pair at a gain of 0.31; the hot read rises to 32.5 mV to restore 10 mV at the comparator	a pull-down gate per node-side line per lane, on a complement select line	2.7×10^{-9}	3.0×10^{-9}	$5.2\times$ less
4×4 , node-side lines track V_y	the selected pair	the same pull-downs, to a per-lane replica of V_y ; the buffer that drives it is unpriced	2.7×10^{-9}	3.0×10^{-9}	$5.2\times$ less
16×1 selector column, pass gates beside the array	the selected pair	sixteen select lines per space instead of eight; no node-side multiplexer	1.4×10^{-9}	1.6×10^{-9}	$9.9\times$ less
16×1 selector column, pass gates stacked along V_y	the selected pair	the same	7.4×10^{-10}	9.1×10^{-10}	$17.2\times$ less

7. Results

Table XIV gives the total at the nominal point and at the stated sensitivities. Changing one nominal parameter at a time gives these DTM-to-CHILL ZEN energy ratios: 6.4 with separate a - and b -side select lines; 4.6 with 16×16 arrays; 6.8 with 45 fJ readout; 5.0 with 0.3 fF/ μ m wire and 0.5 fF gates; 7.5 with 100 nm crossbar pitch; and 6.6 with 0.4 μ m transistor pitch. E_{read} never exceeds 4% of the total at any point in the table.

Ref. [5] estimates cell energy from foundry models for an unnamed advanced process node. The 7 nm-class rows illustrate how such a process could affect our design. Our energy claim uses the nominal row.

8. Bounds on the excluded terms

Rails. The rails run along each lane beside V_y ; their wire capacitance across all 10 080 lanes is 1.4 nF. Assert-

ing a bank’s rails once per read step at that read’s voltage, 128 steps for the g bank and one for each of the others, dissipates 5×10^{-12} J per image, 0.3% of the nominal total.

Leakage. No transistor sits at a memristor, and the select lines have no DC path. The DC paths are the off pass gates of both multiplexers of a pair while its rail is asserted: $2 \times 2 \times 3 = 12$ per space, 2.1×10^7 over the census, at about 5 pA each with at most the read voltage across them. Because the rails are asserted per read, those gates leak only during the read pulses: for pulses of 10 ns per step that is 2.5×10^{-13} J per image, below 10^{-4} of the total. Had the rails been held across an assumed 30 μ s image at 50 mV, the same gates would leak 1.6×10^{-10} J, 8% of the total, which is one reason the rails are pulsed. The image time itself depends on the device’s conductance and readout timing; the energy model does not determine it. The conditional timing scenarios are given in Sec. VB.

Comparator kickback. An illustrative 0.5 fC input

TABLE XIII. The constants of the model and their sources.

constant	value	source
via diameter	20 nm	stated geometry
crossbar pitch d	50 nm	stated geometry; the total is insensitive to it
unit crossbar	4×4	smallest that holds a 16-symbol space; 16×16 as sensitivity
transistor pitch t	$0.20 \mu\text{m}$	28–40 nm-class design rules; $0.06 \mu\text{m}$ for 7 nm-class
pass-gate capacitance C_g	0.3 fF	minimum-size device [25]; 0.1 fF for 7 nm-class
wire capacitance c_w	$0.2 \text{ fF}/\mu\text{m}$	minimum-width metal with neighbors [25]
off-drain capacitance C_d	0.1 fF	same
select supply V_{sel}	0.8 V	core supply; 0.5 V as low-swing sensitivity
read voltage, hot / cold	10 / 50 mV	the operating point of Sec. IV C
comparator + buffer	15 fJ	continuous-time comparator at 200 nA and a 100 nA buffer, both at 0.8 V over the 62.5 ns window; Table XI for the alternatives; 45 fJ as sensitivity
DTM chain	$1.57 \times 10^{-8} \text{ J}$	Ref. [5], Fig. 1, deepest chain, decoded from the figure; their text’s 1.6 T nJ gives 1.28×10^{-8}

charge would move a 131 fF node by about 4 mV, but the first backbone read has only about 12 fF including comparator input capacitance, which gives about 42 mV. A fixed trim cannot be assumed to cancel state-dependent kickback across these loads. Kickback is absent from the emulator and remains a readout-design requirement, not a verified negligible term.

Read disturb. The cold read at 50 mV is two times below the erase threshold and five times below the 0.25 V write threshold; the hot read at 10 mV is ten and twenty-five times below.

9. Basis of comparison

Ref. [5] models its per-sample energy as $E = TK_{\text{mix}}L^2E_{\text{cell}}$ with $E_{\text{cell}} = E_{\text{rng}} + E_{\text{bias}} + E_{\text{comm}} + E_{\text{clock}} \approx 2 \text{ fJ}$ per conditional update, the RNG term measured on a test chip and the rest from foundry models at a node the paper does not name. It excludes weight programming, host communication, and die area. The present model counts the same kinds of term (wire charging, per-cell readout, the analog operation itself) for the same unit of work (one generated image), excludes the same things, and states its process class explicitly. The voltage assumptions favor the DTM: it signals neighbors at

$4V_T \approx 100 \text{ mV}$ and clocks at $5V_T$, while our select lines use the 0.8 V core supply. The 0.5 V row shows the savings from reducing our select voltage. The chain energy used throughout, $1.57 \times 10^{-8} \text{ J}$, is the deepest point of their Fig. 1 decoded from the vector figure (1.961 T nJ at $T = 8$); their text puts the same model at “around 1.6 T nJ,” which is $1.28 \times 10^{-8} \text{ J}$ and makes the nominal ratio 6.4 rather than 7.9. If lane decoding (Sec. VB) is unavailable, a digital 784×226 matrix-vector product adds 1.8–8.9 nJ to the 2.0 nJ estimate, assuming 10–50 fJ per 8-bit multiply-accumulate. The two accountings are therefore comparable in kind, and both are estimates: theirs extrapolated from one fabricated circuit, ours from a stated geometry.

10. Code

The model is one function of the operation count and the constants of Table XIII. It is computed by `chill_zen/energy_geom.py` in the repository <https://github.com/knownm/CHILL-ZEN>, and every row of Table XIV is printed by `experiments/12_energy_model.py` from the banks’ own lane and space counts. The one-level frontier of Fig. 8 and the binary-trained arms are computed by `experiments/13_energy_frontier.py` and `experiments/22_binary_energy.py` from their own operation counts with the same constants. `NUMBERS.md` in the repository lists each number of this appendix beside the script that produces it.

DECLARATIONS

Data and Code Availability

The companion repository contains the final experimental recipe, paper source, saved numerical evidence, and a map from reported quantities to scripts. It is prepared for release at <https://github.com/knownm/CHILL-ZEN>, with the lane emulator at <https://github.com/knownm/ktram-neural-core>. The evaluation of Sec. VC imports the released feature extractor and distance code of Ref. [24] and scores against its shipped reference statistics; we do not redistribute them, and the repository records the commit, the file checksums, and the harness that calls them. The companion repository is released under the MIT license.

Acknowledgments

I thank the authors of Ref. [5] for an energy analysis specific enough to argue with, and the authors of Ref. [24] for publishing the evaluation code that made the argument checkable.

TABLE XIV. The energy terms at each operating point. Every row is printed by the script of Sec. B 10.

operating point	E_{sel}	E_{read}	total (J)	ratio
nominal	1.8×10^{-9}	2.0×10^{-11}	2.0×10^{-9}	7.9 $\times$ less
low-swing select, 0.5 V	7.1×10^{-10}	2.0×10^{-11}	8.8×10^{-10}	17.7 $\times$ less
7 nm-class periphery	6.6×10^{-10}	1.6×10^{-11}	8.2×10^{-10}	19.0 $\times$ less
7 nm-class, 0.5 V	2.6×10^{-10}	1.6×10^{-11}	4.2×10^{-10}	37.0 $\times$ less
pessimistic corner ^a	7.8×10^{-9}	7.9×10^{-11}	8.1×10^{-9}	1.9 $\times$ less

^a 16×16 arrays, separate a - and b -side select lines, 1.0 V.

Every experiment reported here ran on the CPU of my old Intel iMac while I was walking the dogs, asleep, doing chores, or attending to other small businesses. I used AI to run the experiments, gather the data, and draft this paper. I spent many hours trying to get AI to remove its own slop, to the point where it might have been faster to write the whole thing myself. Before release the draft went through several rounds of simulated hostile review, each run with the most capable model available to me at the time, Claude Fable 5.1 and OpenAI Astra-6. They reviewed the paper, code, logs, and source of Ref. [5], then listed gaps, inconsistencies, and misattributions in order of severity. Each round added corrections and more

AI slop, which I removed by hand.

Competing Interests

The author is part of Knowm Inc., which has made and sold memristors and plans to sell products or intellectual property based on technologies described here. The kT-bit, the kT-RAM architecture, and its technology stack are the subject of granted U.S. patents [26–28]; the portfolio is listed at <https://knowm.ai/ip/>.

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